

# Zisen Dai

Berkeley, CA 94704, USA

Website: [zireael-rhena.github.io](https://zireael-rhena.github.io)  
Email: [dzs23@mails.tsinghua.edu.cn](mailto:dzs23@mails.tsinghua.edu.cn)  
[zsai@berkeley.edu](mailto:zsai@berkeley.edu)  
Github: [github.com/Zireael-Rhena](https://github.com/Zireael-Rhena)

*Last Updated: May 27, 2026*

## Education

### **Tsinghua University**

*Bachelor of Science in Physics | GPA: 4.0/4.0 | Rank: 1/28*

August 2023 – Present

*Beijing, China*

### **UC Berkeley**

*Exchange Student in Astrophysics | GPA: 4.0/4.0 | Adviser: Wenbin Lu*

January 2026 – May 2026

*Berkeley, CA, USA*

## Research Experience

### **Department of Astronomy, UC Berkeley**

*Research Assistant (Advised by Wenbin Lu)*

May 2026 – Present

*Berkeley, CA, USA*

- Tracking cutting-edge developments in high-energy transients via weekly group discussions.

### **Institute for Advanced Study, Tsinghua University**

*Research Assistant (Advised by Xuening Bai)*

March 2025 – Present

*Beijing, China*

- Investigating cosmic ray transport dynamics within protoplanetary disk (PPD) winds.
- Developed a Python simulation framework from scratch to model particle transport via the Guiding Center Approximation, analyzing the impact of scattering mechanisms and strengths.

## Project

### **Rosseland-Mean Opacity Computation**

*Calculate the Rosseland-Mean Opacity of Pure Hydrogen Gas under LTE (report, code)*

April 2026

*Berkeley, CA, USA*

- Developed a Python framework to computing the Rosseland-mean opacity of pure hydrogen gas in LTE, and benchmarking it against the LANL/TOPS opacity tables.
- Localized the main opacity discrepancy domains and analyzed the missing physical channels and omitted physics causing the issue.

### **LBGs Selection Based on Photometric Data**

*Machine Learning-Based High-Redshift Target Selection (report, code)*

January 2026

*Beijing, China*

- Developed a LightGBM classifier on CLAUDS photometry, utilizing sample weighting and probability calibration to identify high-redshift ( $z \sim 2 - 3$ ) Lyman Break Galaxies.
- Corrected selection bias via TSR/SSR metrics, achieving 0.995 AUC and 92.0% purity — outperforming the traditional  $u$ -dropout method with a  $3.3\times$  sample size increase.

### **JUNO Optical Simulation & Signal Reconstruction**

*Comprehensive Physics Simulation and Machine Learning Modeling (code)*

August 2025

*Beijing, China*

- Developed a Monte Carlo Python framework to simulate complex photon transport, Fresnel reflection, and emission for 17,612 PMTs.
- Engineered an ultra-high-capacity LightGBM model with multi-center Gaussian attention to reconstruct spatial-temporal Probe functions, reducing temporal integral error to  $\sim 3\%$ .

### **Spectrum and Light Curve Analysis of Supernovae**

*Photometric and Spectroscopic Classification (report, code)*

June 2025

*Beijing, China*

- Classified SN 2025kid as a Type Ia supernova by analyzing multi-band photometric data and spectroscopic absorption features.
- Extracted key parameters (peak magnitude, decline rate  $\Delta m_{15}$ ) of SN 2025fvw via light curve fitting, utilizing the Phillips relation for standardization.

## Awards & Honors

---

|                                                                                                            |             |
|------------------------------------------------------------------------------------------------------------|-------------|
| <b>National Scholarship</b><br><i>Highest Honor for Undergraduates, The Ministry of Education of China</i> | 2024, 2025  |
| <b>Tsinghua Xuetaang Talents Program</b><br><i>Chi-Sun Yeh Class, Elite Physics Track</i>                  | 2023 – 2025 |
| <b>Excellent Academic Performance Scholarship</b><br><i>Tsinghua University</i>                            | 2024, 2025  |
| <b>Basic Disciplines Interdisciplinary Challenge</b><br><i>Third Prize, Tsinghua University</i>            | 2025        |
| <b>Chinese Physics Olympiad (CPhO)</b><br><i>First Prize in Zhejiang</i>                                   | 2022        |

## Selected Coursework

---

**Perfect grades (4.00/4.00)** across all 26 major courses (including **8 A+**, **14 A**)

**Astrophysics:** Relativistic Astrophysics and Cosmology (A+), Statistical Methods in Astrophysics (A+), Observational Astronomy (A+), Radiation Processes in Astronomy, Astrophysics

**Advanced Physics:** Statistical Mechanics (A+), Quantum Mechanics, Electrodynamics, Analytical Mechanics, Atomic and Molecular Physics

**Advanced Mathematics:** Mathematical Physics Equations (A+), Complex Analysis, Linear Algebra for Physical Sciences, Advanced Calculus I & II

**Data:** Big Data in Experimental Physics I & II (A+), Introduction to Probability and Statistics (A+)

## Other Experience

---

|                                                                                                                                                                                 |                                                          |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| <b>Volunteer Peer Tutor</b><br><i>Drop-in Tutoring Center, Tsinghua University</i>                                                                                              | March 2025 – Present<br><i>Beijing, China</i>            |
| <ul style="list-style-type: none"><li>• Providing one-on-one academic support and problem-solving strategies for undergraduates in advanced physics and math courses.</li></ul> |                                                          |
| <b>Student Life Coordinator</b><br><i>Department of Organization, Zhili College</i>                                                                                             | March 2024 – December 2025<br><i>Beijing, China</i>      |
| <ul style="list-style-type: none"><li>• Organized college-wide student events, coordinated group activities, and facilitated team-building initiatives.</li></ul>               |                                                          |
| <b>Academic Representative</b><br><i>Physics Class 32, Zhili College</i>                                                                                                        | September 2023 – September 2025<br><i>Beijing, China</i> |
| <ul style="list-style-type: none"><li>• Liaised between faculty and students to facilitate academic communications and organize study groups.</li></ul>                         |                                                          |